

hw0Rev537

gwf

September 2026

These data are from an agricultural experiment. The response is the dry weight of vegetable matter produced. The factors are cultivar (or a variety of plant that can be cultivated), soil inoculant (a substance put in the soil to encourage growth), and field. There are two cultivars (a and b), 3 soil inoculants (con(trol), dea(d), and liv(e)), and four fields. The data are in a file called 'cultivar.dat'. You can ignore field for now.

1. Perform a standard ANOVA on the 2×3 experiment. Is there an effect due to cultivar? Is there an effect due to inoculant? Is there a significant interaction?
2. Now do a cell means analysis. What are the estimates of the six cell means?
3. Create a contrast matrix to test the effect of cultivars. Is the test significant?
4. Create a contrast matrix to test the effect of inoculants. Is the test significant?
5. Create a contrast matrix for the interactions. (We should have done this first.) Are there significant interactions?
6. Now eliminate all the data in the cultivar b inoculant liv(e) cell. Find the cell means in the remaining 5 cells.
7. Now you need to set one of the degrees of freedom for interaction in the full model analysis to 0. Under that constraint estimate the mean of the cultivar b inoculant liv(e) cell. What is the estimate for that cell? How does it compare to the cell estimate when all the data are present?
8. What is the estimated standard error associated with that cell mean? What is the estimated standard error associated with the cultivar a inoculant liv(e) cell?